SUSPENDED

A CRYOGENIC NIGHTMARE

An advanced-mode route to the best in-game rank

RANK	CASUALTIES	CYCLES	INSTRUCTIONS
1 / 5	**71,000**	**107**	**351,015**

Verified story: Z-machine V3, Release 8, Serial 840521. This fresh-game route reaches the game's best rank (1 of 5), with 71,000 current surface casualties and 0 projected during recovery. The Filtering Computers return to balance in 107 cycles.

How to use this guide

Start a fresh game and enter `ADVANCED` before the first robot order. Follow the stages in order; address each robot explicitly as shown. `ARL` asks all robots to report and advances one cycle. `WAIT` advances three cycles. Keep the access code from the camera view; it is randomly selected for each game.

Run notes

The route consumes 108 player inputs and 351,015 VM instructions. It revealed the code MUMFOO, so Iris entered the second circle and then the first. The game reported no robot death or system-failure event. Whiz was out of communication from the initial report and was not used.

Story fingerprint: SHA-256 7DEEFCD1E4F335DFD7B446F0D6A1DD7E8CD40D71F135FE7C5B788ECBD4749B19

The opcode inventory uses 1-based instruction sequence numbers and hexadecimal byte addresses.

01. Stabilize the surface systems

Choose Advanced mode. Send the three control robots to Weather Control in the shown order, then redirect Poet and Waldo to their assigned controls. The chosen settings hold casualties at the minimum rate while the internal repairs proceed.

```
advanced
sensa, go to weather control
poet, go to weather control
waldo, go to weather control
iris, go to middle supply
auda, go to sleep chamber
poet, go to hydroponics control
waldo, go to transit control
auda, listen
arl
sensa, set second dial to 95
sensa, set first dial to 50
sensa, set third dial to 50
poet, set first lever to 90
poet, set second lever to 60
poet, set third lever to 85
waldo, turn off first switch
waldo, turn off second switch
waldo, turn off third switch
```

02. Open Gamma Repair and restore Iris

Move Waldo, Sensa, and Poet to Middle Supply. While they travel, send Sensa onward to Gamma Repair and Poet to Sub Supply. After the arrival reports, open the large object and turn its switch. Poet retrieves the ramp; use it at the Hallway Junction step so the lower passages are accessible. Waldo repairs Iris with the rough device.

```
waldo, go to middle supply
sensa, go to middle supply
poet, go to middle supply
sensa, go to gamma repair
poet, go to sub supply
arl (repeat seven times)
sensa, examine large object
sensa, turn switch
poet, get ramp
sensa, go to hallway junction
poet, go to hallway junction
waldo, get extension, sixteen-inch cable, red ic, yellow ic and rough object
waldo, wear extension
waldo, open smooth panel
waldo, replace rough device with rough object
waldo, close smooth panel
poet, put ramp by step
poet, go to hallway corner
sensa, go to hallway corner
sensa, get ramp
waldo, drop extension and rough device
```

03. Collect the cutter and camera; open the machine

Send Waldo and Iris north to Main Supply. Sensa moves through the Sleep Chamber to Small Supply for the cutter, then carries the ramp back to the Sloping Corridor. Poet uses the carriage at Hallway End to reach the Biology Laboratory. Repair the machine with both ICs, then use its fuse to exchange the damaged fourteen-inch cable for the sixteen-inch cable.

```
waldo, north
iris, north
poet, go to hallway end
sensa, north
sensa, go to small supply room
iris, get burned and fried
waldo, put red ic in red socket
sensa, drop ramp
sensa, stand on ramp
sensa, get cutter
sensa, get ramp
sensa, go to sloping corridor
poet, enter carriage
poet, get out
poet, go to biology laboratory
waldo, put yellow ic in yellow socket
poet, get camera
poet, go to vehicle debark area
sensa, put ramp by step
waldo, push button
poet, enter carriage
poet, get out
sensa, go to southeast junction
poet, go to southeast junction
waldo, get fuse
waldo, replace fourteen-inch cable with sixteen-inch cable
waldo, insert fuse in machine
waldo, go to gamma repair
arl (repeat four times)
poet, give camera to sensa
sensa, go to gamma repair
poet, go to hydroponics control
arl (repeat four times)
```

04. Free Fred and redirect the intruders

At Gamma Repair, have Waldo and Sensa move the broken device together. Sensa hands Waldo her camera and tool; Waldo cuts Fred's twelve-inch cable free. Auda takes the intruders' tool bag and leads them to Maintenance Access. Listen until they discover and stop the acid leak before sending Waldo through the lower channel route.

```
both waldo and sensa, get broken device
sensa, give all to waldo
waldo, cut twelve-inch cable with cutter
auda, get bag
auda, go to maintenance access
waldo, get twelve-inch cable
waldo, go to short corridor
sensa, go to weather control
```

05. Balance the computers and read the reset code

Poet resets the hydroponics controls and returns to Weather Control. Waldo replaces the damaged Primary and Secondary Channel cables, then plugs in the camera to read the code. For this supplied story run the code is MUMFOO: press MUM (second circle), followed by FOO (first circle).

```
poet, set second lever to 45
waldo, go to primary channel
poet, set third lever to 70
poet, go to weather control
wait (once)
poet, set first dial to 55
poet, set third dial to 55
waldo, replace four-inch cable with twelve-inch cable
waldo, south
waldo, south
waldo, replace nine-inch cable with fourteen-inch cable
waldo, plug camera into hole
waldo, point camera at sign
iris, press second circle
iris, press first circle
```

Completion check: the Filtering Computers report that all systems are returning to normal, surface life is in recovery mode, and total recovery is projected in 10 cycles. The game reports rank 1 of 5 and successful completion in 107 cycles.

Executed opcode inventory

54 unique opcodes were observed across 351,015 instructions in the fresh-game route. Counts include all VM instructions. Opcode numbers are hexadecimal; the instruction form distinguishes encodings. PCs are byte addresses.

Form	Opcode	Mnemonic	Count	First seq.	First PC
Long	0x0D	store	21,020	1	0x0060E9
Variable	0x00	call	24,097	5	0x0060F5
Long	0x14	add	7,897	7	0x006055
Long	0x01	je	53,792	9	0x00605D
Long	0x15	sub	8,968	10	0x006061
Short	0x00	jz	55,595	11	0x006065
Variable	0x01	storew	6,504	13	0x006070
Short	0x0B	ret	1,639	14	0x006075
Long	0x0F	loadw	29,222	23	0x006077
Short	0x0C	jump	14,944	26	0x006085
Long	0x0A	test_attr	14,979	493	0x00619D
Short	0x02	print	834	495	0x008345
Variable	0x09	and	1	497	0x0083DA
Variable	0x06	print_num	17	498	0x0083E0
Long	0x05	inc_chk	14,673	500	0x0083F2
Long	0x10	loadb	6,907	501	0x0083F9
Variable	0x05	print_char	6	502	0x0083FD
Short	0x0B	new_line	372	526	0x008403
Short	0x00	rtrue	2,159	527	0x008404
Long	0x11	get_prop	3,795	534	0x0061F0
Long	0x03	jg	6,207	536	0x0061F8
Short	0x03	get_parent	2,028	539	0x006209
Long	0x06	jin	709	540	0x00620C
Variable	0x08	push	1,592	541	0x006210
Short	0x05	inc	3,054	552	0x00A4F8
Variable	0x01	je	8,704	579	0x00657A
Short	0x01	rfalse	6,061	590	0x006685
Short	0x0D	print_paddr	577	626	0x00AF73
Short	0x0A	print_obj	235	632	0x00A618
Short	0x03	print_ret	151	633	0x00A61A
Short	0x08	ret_popped	1,578	1,145	0x00A542
Long	0x09	and	981	1,251	0x00683D
Variable	0x04	sread	108	1,267	0x00685F
Long	0x04	dec_chk	1,332	1,275	0x006894
Long	0x07	test	5,218	1,285	0x006B1D
Long	0x16	mul	885	1,301	0x0069B5
Variable	0x02	storeb	296	1,304	0x0069C1
Long	0x02	jl	7,232	1,328	0x006F8A
Long	0x0B	set_attr	46	1,668	0x00AD99
Variable	0x0D	store	386	2,433	0x0068E2
Short	0x02	get_child	6,476	2,539	0x007A21

Form	Opcode	Mnemonic	Count	First seq.	First PC
Long	0x12	get_prop_addr	11,793	2,542	0x007A38
Short	0x04	get_prop_len	5,941	2,548	0x007D58
Long	0x17	div	5,699	2,549	0x007D5B
Short	0x01	get_sibling	5,867	2,573	0x007AA6
Variable	0x03	jg	141	7,012	0x00BC8A
Long	0x0E	insert_obj	241	7,195	0x00D977
Long	0x0C	clear_attr	40	65,334	0x008573
Variable	0x15	sub	9	167,067	0x00DBE0
Short	0x07	print_addr	2	196,029	0x007338
Variable	0x07	random	2	207,049	0x005F11
Variable	0x17	div	1	350,982	0x00AC47
Variable	0x18	mod	1	350,983	0x00AC4D
Short	0x0A	quit	1	351,015	0x00BC46

Replay completion: Halted; output SHA-256 FF2AC0B25EE7377F7720CE1A8B30E51B5BF18FB231D5239944FD048134FF8653. The deterministic route and machine-readable opcode trace are stored locally under .agent/regression-routes/ and are intentionally ignored by Git.